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unitary operator U

$$UU^\dagger = U^\dagger U = \mathbb{1}$$

$$U|\psi\rangle = e^{i2\pi\phi} |\psi\rangle$$

True if $|\psi\rangle$ is an eigenstate of U

$$U|\psi\rangle = \lambda |\psi\rangle$$

$$|\lambda| = 1; \quad \lambda = e^{i\theta}$$

for some real θ

$$\text{if } a|4\rangle = \lambda|4\rangle$$

$$\|a|4\rangle\|_2^2 =$$

$$\langle 4|a a^\dagger|4\rangle = \langle 4|4\rangle$$

But also

$$\|a|4\rangle\|_2^2 = \|\lambda|4\rangle\|_2^2$$

$$= \lambda^2 \underbrace{\langle 4|4\rangle}_1 \Rightarrow |\lambda| = 1$$

since $|\lambda| = 1$, every eigenvalue lies on the unit circle

$$\lambda = e^{i\theta}$$

The phase θ 's periodic modulo 2π ($e^{i(\theta + 2\pi)} = e^{i\theta}$)

$$\theta = 2\pi\phi; \phi \in [0, 1)$$

$$\lambda = e^{2\pi i\phi}$$

what about a General state?

Let $\{|u_j\rangle\}$ be an eigen-basis of U .

For a general state

$$|\psi\rangle = \sum_j c_j |u_j\rangle$$

$$U|\psi\rangle = \sum_j c_j e^{2\pi i \phi_j} |u_j\rangle$$

Spectral theorem (in finite dim)

$$U |u_j\rangle = e^{i\theta_j} |u_j\rangle$$

$$U = \sum_j e^{i\theta_j} |u_j\rangle \langle u_j|$$

$$= \sum_j e^{2\pi i \phi_j} |u_j\rangle \langle u_j|$$

$$U = \sum_j \lambda_j P_j : \begin{cases} P_j = |u_j\rangle \langle u_j| \\ \lambda_j = e^{i\theta_j} \end{cases}$$

In our case we are interested in the eigenvalues of an operator A , which is not necessarily unitary, but we assume it is Hermitian

$$A|\psi\rangle = a|\psi\rangle$$

Time evolution $-iAt$

$$U(t) = e$$

$$\text{if } A|\psi_i\rangle = a_i|\psi_i\rangle$$

$$U(t) |\psi_i\rangle = e^{-i a_i t} |\psi_i\rangle$$

$$U(t) = e^{-i A t} =$$

$$\sum_j e^{-i a_j t} |\psi_j\rangle \langle \psi_j|$$

GPE begins with the relation

$$U |\psi_j\rangle = e^{2\pi i \phi_j} |\psi_j\rangle$$

The goal is to estimate ϕ_j

why QPE works-

The ingredients are

- an eigenstate $|u_j\rangle$

- controlled powers of rotation gates

- an inverse QFT

$$U^{2^k} |u_j\rangle = e^{2\pi i 2^k \phi_j} |u_j\rangle$$

if the input is not an eigenstate, then

$$|\psi\rangle = \sum c_j |\psi_j\rangle,$$

then the QPE produces

$$\sum c_j |\hat{E}_j\rangle |\psi_j\rangle$$

so that the measurement outcome samples one of the eigenphases with a probability $|c_j|^2$

$$u = e^{iAt}$$

$$e^{iAt} |u_j\rangle = e^{i\lambda_j \cdot t} |u_j\rangle$$

$$\phi_j = \frac{\lambda_j \cdot t}{2\pi}$$

aside: this is also the basis of the HHL algo

$$Ax = b \quad (x = A^{-1}b)$$

QPE

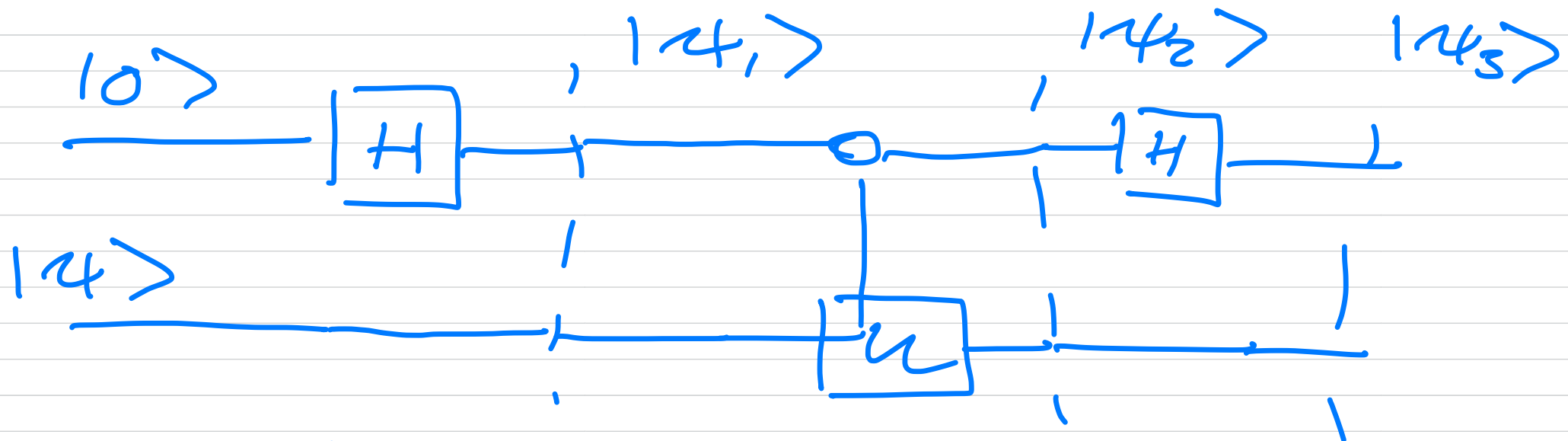
Basic recipe

- initialize n -bits in $|0\rangle$,

$$|0\rangle \otimes |0\rangle \dots \otimes |0\rangle = |0\rangle^{\otimes n}$$

(ancilla qubits), control register

- second register (system) contains all the qubits which are needed to define the state $|u\rangle$



$$|0\rangle \otimes |4\rangle$$

$$|0\rangle |4\rangle$$

$$|4_1\rangle = \frac{1}{\sqrt{2}} (|0\rangle \otimes |4\rangle + |1\rangle \otimes |4\rangle)$$

control - U

$$|4_2\rangle = \frac{1}{\sqrt{2}} (|0\rangle \otimes |4\rangle + e^{2\pi i 0. \phi_0} |1\rangle \otimes |4\rangle)$$

$$0. \phi_0 = \phi_0 \cdot \frac{1}{2}$$

$$\phi_0 = \{0, 1\}$$

$$= \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i' C \cdot \phi_0} |1\rangle \right) \otimes |\psi\rangle$$

Hadamard on $|\psi_2\rangle$

$$H |\psi_2\rangle = |\psi_3\rangle$$

$$= \frac{1}{2} \left((1 + e^{2\pi i' C \cdot \phi_0}) |0\rangle + (1 - e^{2\pi i' C \cdot \phi_0}) |1\rangle \right) \otimes |\psi\rangle$$

Measure on $|0\rangle$ or $|1\rangle$

ϕ_0 is either 0 or 1

if $\phi_0 = 0$, then

$$|u_3\rangle = |0\rangle$$

if $\phi_0 = 1$, then $|u_3\rangle = |1\rangle$

if we measure with certainty
 $|0\rangle$ or $|1\rangle$, then $\phi_0 = 0$

or $1, \frac{1}{2} = 0.5$. we know
the eigenvalue

$$\lambda = e^{2\pi i \phi}$$

with one qubit only, we
have either 0 or 0.5

with two qubits

$$C, \phi_0 \phi_1 = \phi_0 \cdot \frac{1}{2} + \phi_1 \cdot \frac{1}{4}$$

$$\phi_0, \phi_1 = \{0, 1\}$$

$$\phi = \left\{ 0, \frac{1}{2}, \frac{1}{4}, \frac{3}{4} \right\}$$